

OS Skill : 1

```
gssgvy@gyanadeep:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [417 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.2 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [349 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [81.0 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [67.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [367 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [70.8 kB]
Fetched 3159 kB in 5s (655 kB/s)
74 packages can be upgraded. Run 'apt list --upgradable' to see them.
gssgvy@gyanadeep:~$ sudo apt install gcc git make valgrind -y
gcc is already the newest version (4:15.2.0-8ubuntu1).
git set to manually installed.
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
make is already the newest version (4.4.1-3).
make set to manually installed.
Upgrading:
  libc-bin libc-dev-bin libc-gconv-modules-extra libc6 libc6-dev locales

Installing:
  valgrind

Installing dependencies:
  gdb libbabeltracec libdebuginfod1t64 libipt2 libsource-highlight-common libsource-highlight4t64
  libbabeltracel libdebuginfod-common libdw1t64 libsource-highlight-common

Suggested packages:
  gdb-doc gdbserver valgrind-mpi kcachegrind

Summary:
  Upgrading: 6, Installing: 19, Removing: 0, Not Upgrading: 68
  Download size: 27.1 MB / 37.8 MB
  Space needed: 111 MB / 1023 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltracel amd64 1:5.11-5build1 [165 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 gdb amd64 17.1-2ubuntu1 [4173 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dev amd64 2.43-2ubuntu2.3 [6407 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 valgrind amd64 1:3.26.0-0ubuntu1 [15.7 MB]
Fetched 27.1 MB in 18s (1479 kB/s)
Preconfiguring packages ...
(Reading database ... 48264 files and directories currently installed.)
Preparing to unpack .../libc-dev-bin_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc-dev-bin (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...
Preparing to unpack .../libc6-dev_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc6-dev:amd64 (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...
Preparing to unpack .../libc-gconv-modules-extra_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...
Setting up libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.3) ...
(Reading database ... 48264 files and directories currently installed.)
Preparing to unpack .../libc6_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc6:amd64 (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...
Setting up libc6:amd64 (2.43-2ubuntu2.3) ...
(Reading database ... 48264 files and directories currently installed.)
Preparing to unpack .../libc-bin_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc-bin (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...
Setting up libc-bin (2.43-2ubuntu2.3) ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 48264 files and directories currently installed.)
Preparing to unpack .../00-libdebuginfod-common_0.194-4_all.deb ...
Unpacking libdebuginfod-common (0.194-4) ...
Preparing to unpack .../01-locales_2.43-2ubuntu2.3_all.deb ...
```

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Setting up libc-bin (2.43-2ubuntu2.3) ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 48264 files and directories currently installed.)
Preparing to unpack .../00-libdebuginfod-common_0.194-4_all.deb ...
Unpacking libdebuginfod-common (0.194-4) ...
Preparing to unpack .../01-locales_2.43-2ubuntu2.3_all.deb ...
Unpacking locales (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...
Selecting previously unselected package libdw1t64:amd64.
Preparing to unpack .../02-libdw1t64_0.194-4_amd64.deb ...
Unpacking libdw1t64:amd64 (0.194-4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../03-libbabeltrace1_1.5.11-5build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../04-libdebuginfod1t64_0.194-4_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../05-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../06-libsource-highlight-common_3.1.9-4.3build2_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build2) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../07-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../08-gdb_17.1-2ubuntu1_amd64.deb ...
Unpacking gdb (17.1-2ubuntu1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../09-libc6-dbg_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.3) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../10-valgrind_1%3a3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up locales (2.43-2ubuntu2.3) ...
Generating locales (this might take a while)...
Generation complete.
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.3) ...
Setting up libdw1t64:amd64 (0.194-4) ...

```

```

Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up libc-dev-bin (2.43-2ubuntu2.3) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Setting up libc6-dev:amd64 (2.43-2ubuntu2.3) ...
Processing triggers for systemd (259.5-0ubuntu3) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
gssggy@Gyanadeep:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

gssggy@Gyanadeep:~$ git --version
git version 2.53.0
gssggy@Gyanadeep:~$ valgrind --version
valgrind-3.26.0
gssggy@Gyanadeep:~$ mkdir ossp
gssggy@Gyanadeep:~$ cd ossp
gssggy@Gyanadeep:~/ossp$ mkdir src include build
gssggy@Gyanadeep:~/ossp$ ls
build include src
gssggy@Gyanadeep:~/ossp$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"

```

```
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/gssgyy/oss/oss/.git/
gssgyy@Gyanadeep:~/oss$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
gssgyy@Gyanadeep:~/oss$ touch README.md
gssgyy@Gyanadeep:~/oss$ git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'
hint: Disable this message with "git config set advice.addEmptyPathsSpec false"
gssgyy@Gyanadeep:~/oss$ git commit -m "Initial Initial Project setup"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <gssgyy@Gyanadeep.localdomain>) not allowed
gssgyy@Gyanadeep:~/oss$ cs src
Command 'cs' not found, but can be installed with:
sudo apt install csound
```

On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

```
gssggy@Gyanadeep:~/ossp$ touch README.md
```

```
gssggy@Gyanadeep:~/ossp$ git add
```

Nothing specified, nothing added.

hint: Maybe you wanted to say 'git add .'?

hint: Disable this message with "git config set advice.addEmptyPathsSpec false"

```
gssggy@Gyanadeep:~/ossp$ git commit -m "Initial Initial Project setup"
```

Author identity unknown

*** Please tell me who you are.

Run

```
git config --global user.email "you@example.com"
```

```
git config --global user.name "Your Name"
```

to set your account's default identity.

Omit --global to set the identity only in this repository.

fatal: empty ident name (for <gssggy@Gyanadeep.localdomain>) not allowed

```
gssggy@Gyanadeep:~/ossp$ cs src
```

Command 'cs' not found, but can be installed with:

```
sudo apt install csound
```

```
gssggy@Gyanadeep:~/ossp$ nano.c
```

Command 'nano.c' not found, did you mean:

command 'nanoc' from deb nanoc (4.14.6-1)

Try: sudo apt install <deb name>

```
gssggy@Gyanadeep:~/ossp$ nano main.c
```

```
gssggy@Gyanadeep:~/ossp$ gcc main.c -o ../build/main
```

/usr/bin/x86_64-linux-gnu-ld.bfd: cannot open output file ../build/main: No such file or directory

collect2: error: ld returned 1 exit status

```
gssggy@Gyanadeep:~/ossp$ ../build/main
```

-bash: ../build/main: No such file or directory

```
gssggy@Gyanadeep:~/ossp$ nano fork_demo.c
```

```
gssggy@Gyanadeep:~/ossp$ gcc fork_demo.c -o fork_demo
```

```
gssggy@Gyanadeep:~/ossp$ ./fork_demo
```

Parent Process

to set your account's default identity.

Omit --global to set the identity only in this repository.

fatal: empty ident name (for <gssggy@Gyanadeep.localdomain>) not allowed

```
gssggy@Gyanadeep:~/ossp$ cs src
```

Command 'cs' not found, but can be installed with:

```
sudo apt install csound
```

```
gssggy@Gyanadeep:~/ossp$ nano.c
```

Command 'nano.c' not found, did you mean:

command 'nanoc' from deb nanoc (4.14.6-1)

Try: sudo apt install <deb name>

```
gssggy@Gyanadeep:~/ossp$ nano main.c
```

```
gssggy@Gyanadeep:~/ossp$ gcc main.c -o ../build/main
```

/usr/bin/x86_64-linux-gnu-ld.bfd: cannot open output file ../build/main: No such file or directory

collect2: error: ld returned 1 exit status

```
gssggy@Gyanadeep:~/ossp$ ../build/main
```

-bash: ../build/main: No such file or directory

```
gssggy@Gyanadeep:~/ossp$ nano fork_demo.c
```

```
gssggy@Gyanadeep:~/ossp$ gcc fork_demo.c -o fork_demo
```

```
gssggy@Gyanadeep:~/ossp$ ./fork_demo
```

Parent Process

Parent PID: 1967

Child Process

Child PID: 1968

```
gssggy@Gyanadeep:~/ossp$ nano exec_demo.c
```

```
gssggy@Gyanadeep:~/ossp$ gcc exec_demo.c -o exec_demo
```

```
gssggy@Gyanadeep:~/ossp$ ./exec_demo
```

Before exec

total 56

```
-rw-r--r-- 1 gssggy gssggy 0 Aug 24 05:21 README.md
drwxr-xr-x 2 gssggy gssggy 4096 Aug 24 05:20 build
-rwxr-xr-x 1 gssggy gssggy 16000 Aug 24 05:25 exec_demo
-rw-r--r-- 1 gssggy gssggy 167 Aug 24 05:25 exec_demo.c
-rwxr-xr-x 1 gssggy gssggy 16080 Aug 24 05:24 fork_demo
-rw-r--r-- 1 gssggy gssggy 364 Aug 24 05:24 fork_demo.c
drwxr-xr-x 2 gssggy gssggy 4096 Aug 24 05:20 include
-rw-r--r-- 1 gssggy gssggy 79 Aug 24 05:23 main.c
drwxr-xr-x 2 gssggy gssggy 4096 Aug 24 05:20 src
gssggy@Gyanadeep:~/ossp$ |
```